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ABSTRACT

An airplane or aeroplane (informally plane) is a powered, fixed-wing aircraft that is propelled forward by thrust from a jet engine, propeller or rocket engine. Airplanes come in a variety of sizes, shapes, and wing configurations. The broad spectrum of uses for airplanes includes recreation, transportation of goods and people, military, and research. Worldwide, commercial aviation transports more than four billion passengers annually on airliners and transports more than 200 billion tonnes of cargo annually, which is less than 1% of the world's cargo movement. Most airplanes are flown by a pilot on board the aircraft, but some are designed to be remotely or computer-controlled. This report is a summary of the electrical, electronic and the radio communication wings of a modern commercial aircraft under operation with Air India Limited.

ACKNOWLEDGEMENT

I would like to take this opportunity to express my gratitude towards Air India Limited for providing me with this wonderful opportunity of interning with it. I would also like to thank all the respective and respected teachers of the various departments who shared their invaluable wisdom with me and my group. I am especially grateful towards the different shop in-charges who were always there to guide us during our internship and helped us in getting to know the subject matter more proficiently. Finally, I am appreciative of my family and friends for helping me out throughout this industrial training.

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LIST OF ABBREVIATIONS

CDI	**Course Director Indicator**
FDI	**Flight Director Indicator**
ILS	**Instrument Landing System**
SSFDR	**Solid State Flight Data Recorder**
CVR	**Cockpit Voice Recorder**
ELT	**Emergency Location Transmitter**

INTRODUCTION

An aircraft is an assimilation of multiple instruments. It is of paramount importance that these instruments are checked and maintained on a regular basis. The instruments can be varying in operation. There are flight instruments which are used in controlling the aircraft`s flight altitude. In this respect both the Course Director Indicator (CDI) and the Flight Director Indicator (FDI) play vital roles. These instruments when used alongside the Instrument Landing System (ILS) and its localizer and glidescope, can help in automatic aircraft landing. There is a plethora of other instruments including engine instruments, navigation instruments and electrical instruments. The radio instruments used help in air traffic control by allowing communication between the base station and the airplane as well as between two different aircrafts. Different communication and navigational instruments used in the aircraft have been discussed. Another important feature with which an aeroplane is equipped includes the crash detection system. The black boxes- Solid State Flight Data Recorder (SSFDR) and the Cockpit Voice Recorder (CVR) have an Emergency Location Transmitter (ELT) which starts transmitting the crash location information upon detecting an aircraft crash. Apart from that there is also an underwater location beacon which transmits under water and is helpful in water crash detection. The electrical shop deals with the various components of the Auxiliary Power Unit(APU) wherein a generator is coupled with the aircraft engine on both the wings and generates electricity which acts as a backup in case all other electrical systems fail and powers only the most important components in case of an emergency.

INSTRUMENT OVERHAUL

2.1 COURSE DIRECTOR INDICATOR (CDI)

Course Indicator is an aircraft navigation instrument which displays a pictorial plane view of an aircraft with respect to magnetic north, to the selected course and selected heading. The magnetic heading, selected heading, selected course and heading deviation are read against a servo-driven azimuth card. Selected course and distance information is displayed.

Indicators: The azimuth card repeats heading information received from gyro-stabilised magnetic compass. The magnetic heading of the aircraft is read under lubber line.

Miniature Aircraft: The miniature aircraft is fixed at the center of the instrument and is aligned to point to the lubber line, therefore pointing to the magnetic heading of the aircraft.

Compass warning shutter: It is located at the top of the instrument which is retracted to display the compass flag when the heading power or compass monitor is lost.

Heading marker: It indicates the selected heading against the azimuth card. The displacement of the heading marker from the lubber line shows the direction and magnitude of deviation from the selected heading.

Course, arrow and display: It identifies the selected VOR or localizer course. Once set the course arrow will rotate with the azimuth card and give a continuous reading of the selected course.

Course Bar (lateral deviation bar): It represents the selected VOR or localizer course. If the course bar is aligned with the course and reciprocal course arrows, the aircraft is on the selected course.

To-From-Arrow: This arrow points towards the VOR station, therefore the pilot is able to determine if his aircraft is flying to or from the station.

VOR-LOC flag: This flag is located on the right side of the instrument face. It becomes visible when the VOR or localizer signal is lost or is too weak to provide reliable information.

Glide-slope Pointer: This flag is located on the left side of the instrument face. It is read against a graduated scale to indicate the location of the glide path with respect to the aircraft.

Glide slope flag: It is located on the left side of the instrument face, it becomes visible and covers the glide slope pointer and scale, when the glide slope signal is lost or becomes too weak to provide reliable information.

Distance display: Units, tenths and hundreds distance data in the form of three wire inputs is applied to the stators of three synchro receivers. The distance counters then display Doppler or DME information.

2.2 FLIGHT DIRECTOR INDICATOR (FDI)

The FDI represents pictorial display to the pilot of the aircraft and flight director information.

Table 2.1 Flight Director Indicator

CHARACTERISTICS	SPECIFICATIONS
Size	Bezel size: 5 inch wide by 5.250inch high. Case size:4.875inch wide by 5.125inch high by 7.50inch long
Weight	7.51lb(3.4kg) max
Colour	Case: Grey Attitude display sky: Blue
Ambient Temperature	Operation:-30degree centigrade to 55 degree centigrade. Storage:-65degree centigrade to 70

	degree centigrade Test:-Normal factory ambient
Humidity	Relative humidity of 95% at +2degree centigrade to 32 degree centigrade Absolute humidity:95% of relative humidity at +32 degree centigrade from +32 degree centigrade to +72 degree centigrade.
Attitude	-1000 to +50000 feet

Fig 2.1 Cockpit Instruments System

RADIO OVERHAUL

Aviation communication refers to the conversing of two or more aircraft. Aircraft are constructed in such a way that makes it very difficult to see beyond what is directly in front of them. As safety is a primary focus in aviation, communication methods such as wireless radio are an effective way for aircraft to communicate with the necessary personnel. Aviation is an international industry and as a result involves multiple languages. However, as deemed by the International Civil Aviation Organization (ICAO), English is the official language of aviation. The industry considers that some pilots may not be fluent English speakers and as a result pilots are obligated to participate in an English proficiency test.

Aviation communication is the means by which aircraft crews connect with other aircraft and people on the ground to relay information. Aviation communication is a crucial component pertaining to the successful functionality of aircraft movement both on the ground and in the air. Increased communication reduces the risk of an accident.

Aviation was roaring during the Jet Age and as a result, communication technologies needed to be developed. This was initially seen as a very difficult task: ground controls used visual aids to provide signals to pilots in the air. With the advent of portable radios small enough to be placed in planes, pilots were able to communicate with people on the ground. With later developments, pilots were then able to converse air-to-ground and air-to-air. Today, aviation communication relies heavily on the use of many systems. Planes are outfitted with the newest radio and GPS systems, as well as Internet and video capabilities.

3.1 ITU RADIO SPECTRUM ALLOCATIONS

The allocation of radio spectrum is defined by the International Telecommunications Union (ITU) and relates the use of a frequency to a specific service. In the case of civil aviation there are separate ITU allocations for communications, navigation, and surveillance. Such differentiation between functions corresponds to the safety requirements for Air Traffic Control. The ITU has assigned frequencies for use by aircraft analog voice dialogue in parts of the "High Frequency" (3-30 MHz) band and in the 118-137 MHz section of the wider "Very High Frequency" range. Aircraft can use radios operating in the HF radio band for long-range communications as the signals are reflected by the ionosphere. Unfortunately when using HF the link audio quality is very poor due to the long propagation of the wave. Aircraft can use radios operating in the VHF band to communicate with other radios in line-of-sight coverage. These signals do not reflect off the ionosphere or penetrate obstacles such as mountains or buildings. The advantage of VHF over HF is that the link quality is much better and there is greater reuse of the frequency channel. The use of the word "analog" in relation to voice radio communications means that the changes in the sound of the voice are converted by the transmitter into corresponding variations in the radio signal and converted back by the receiver.

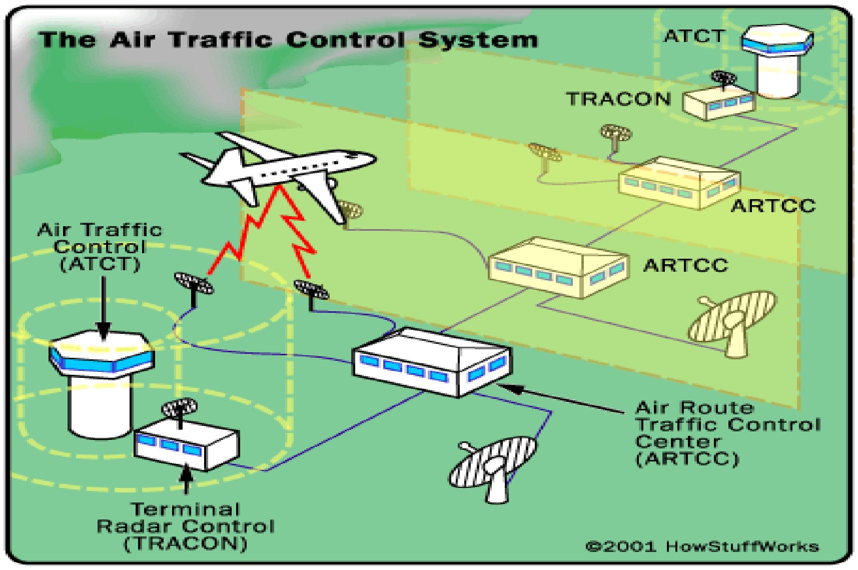

FIG 3.1 WORKING OF RADIO COMMUNICATION

3.2 COCKPIT VOICE RECORDER:

Cockpit Voice Recorder system consists of recorder and externally mounted control unit or externally mounted e-amplifier with remote microphone.

The voice recorder has an aluminium chassis, cover painted international orange with two reflective stripes located on a stainless steel or titanium Crash Survival Memory Unit (C.S.M.U). The recorder weighs in pounds with an underwater locator device mounted on the front of the CSMU.

The functions of the CVR are summarised as follows:

a) Power, control and monitor signal interface characteristics

b) Voice data interface characteristics

It has 4 audio inputs which are connected as follows:-

Channel 1- cockpit spare audio input

Channel 2-Co-pilots audio, boom, mask, hand-held microphone input.

Channel 3- Pilot's Audio, Boom, mask, hand-held microphone unit.

Channel 4-Cockpit's Area Microphone (CAM) input.

c) Data Recording Storage

d) Audio Monitoring

e) Crash Survival Memory Unit

3.3 EMERGENCY LOCATOR TRANSMITTER

The ELT is a self contained transmitter that is designed to help locate an airplane after crash. The ELT is battery powered and is automatically turned on by crash forces. It has a floatation device and a flexible antenna. The transmitter is activated by an acceleration operator switch when a rapid deceleration force is applied along the longitudinal axis of the aircraft. It transmits a digital distress signal to satellites. The signal is transmitted on the 406MHz frequency and used for accurate localization and identification. It also transmits other homing signals at 121.5MHz and 243MHz frequencies for an easier final approach in emergency conditions.

There are two types of ELTs: Crash activated and water activated.

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ELECTRIC OVERHAUL

An Aircraft Electrical System is a self contained network of components that generate, transmit, distribute, utilize and store electrical energy.

An electrical system is an integral and essential component of all but the most simplistic of aircraft designs. The electrical system capacity and complexity varies tremendously between a light, piston powered, single engine GA aircraft and a modern, multiengine commercial jet aircraft. However, the electrical system for aircraft at both ends of the complexity spectrum share many of the same basic components.

All aircraft electrical systems have components with the ability to generate electricity. Depending upon the aircraft, generators or alternators are used to produce electricity. These are usually engine driven but may also be powered by an APU, a hydraulic motor or a Ram Air Turbine (RAT) (RAT). Generator output is normally 115-120V/400Hz AC, 28V DC or 14V DC. Power from the generator may be used without modification or it may be routed through transformers, rectifiers or inverters to change the voltage or type of current.

The generator output will normally be directed to one or more distribution bus. Individual components are powered from the bus with circuit protection in the form of a Circuit Breaker or fuse incorporated into the wiring.

The generator output is also used to charge the aircraft battery(s). Batteries are usually either of lead-acid or NICAD types but lithium batteries are becoming more and more common. They are used for both aircraft startup and as an emergency source of power in the event of a generation or distribution system failure.

4.1 BASIC AIRCRAFT ELECTRICAL SYSTEMS

Some very simple single engine aircraft do not have an electrical system installed. The piston engine is equipped with a Magneto ignition system, which is self powering, and the fuel tank is situated such that it gravity feeds the engine. The aircraft is started by means of a flywheel and crank arrangement or by "hand-propping" the engine.

If an electric starter, lights, electric flight instruments, navigation aids or radios are desired, an electrical system becomes a necessity. In most cases, the system will be DC powered using a single distribution bus, a single battery and a single engine driven generator or alternator. Provisions, in the form of an on/off switch, will be incorporated to allow the battery to be isolated from the bus and for the generator/alternator to be isolated from the bus. An ammeter, load meter or warning light will also be incorporated to provide an indication of charging system failure. Electrical components will be wired to the bus-bar incorporating either circuit breakers or fuses for circuit protection. Provisions may be provided to allow an.

4.2 ADVANCED AIRCRAFT ELECTRICAL SYSTEMS

More sophisticated electrical systems are usually multiple voltage systems using a combination of AC and DC buses to power various aircraft components. Primary power generation is normally AC with one or more Transformer Rectifier Unit (TRU) providing conversion to DC voltage to power the DC busses. Secondary AC generation from an APU is usually provided for use on the ground when engines are not running and for airborne use in the event of component failure. Tertiary generation in the form of a hydraulic motor or a RAT may also be incorporated into the system to provide redundancy in the event of multiple failures. Essential AC and DC components are wired to specific buses and special provisions are made to provide power to these buses under almost all failure situations. In the event that all AC power generation is lost, a static inverter is included in the system so the Essential AC bus can be powered from the aircraft batteries.

Robust system monitoring and failure warning provisions are incorporated into the electrical system and these are presented to the pilots when appropriate. Warnings may include, but are not limited to, generator malfunction/failure, TRU failure, battery failure, bus fault/failure and circuit breaker monitoring. The manufacturer will also provide detailed electrical system isolation procedures to be utilized in the event of an electrical fire.

In compliance with applicable regulations, components such as Aircraft Emergency Floor Path Illumination have their own backup power supplies and will function even in the event of a complete electrical system failure.

Provisions are virtually always provided for connecting the aircraft electrical system to a fixed or mobile Ground Power Unit (GPU).

4.3 TEMPERATURE CONTROL THERMOSTAT (TCT)

The TCT has a rod which senses the temperature of the pneumatic duct above it which carries pre-cooled air. If the temperature exceeds 200°C then it partially opens the fan air valve to increase the flow of cool air into the pre-cooler. This will reduce the temperature of the hot air flowing into the pre-cooler. If the temperature exceeds 260°C then the TCT will open the fan air valve completely. The TCT can regulate the fan air valve. This means that the TCT can open or close the fan air valve to any degree as per situation. Movement of fan air valve is related to differential thermal expansion between the stainless steel and the invar rod.

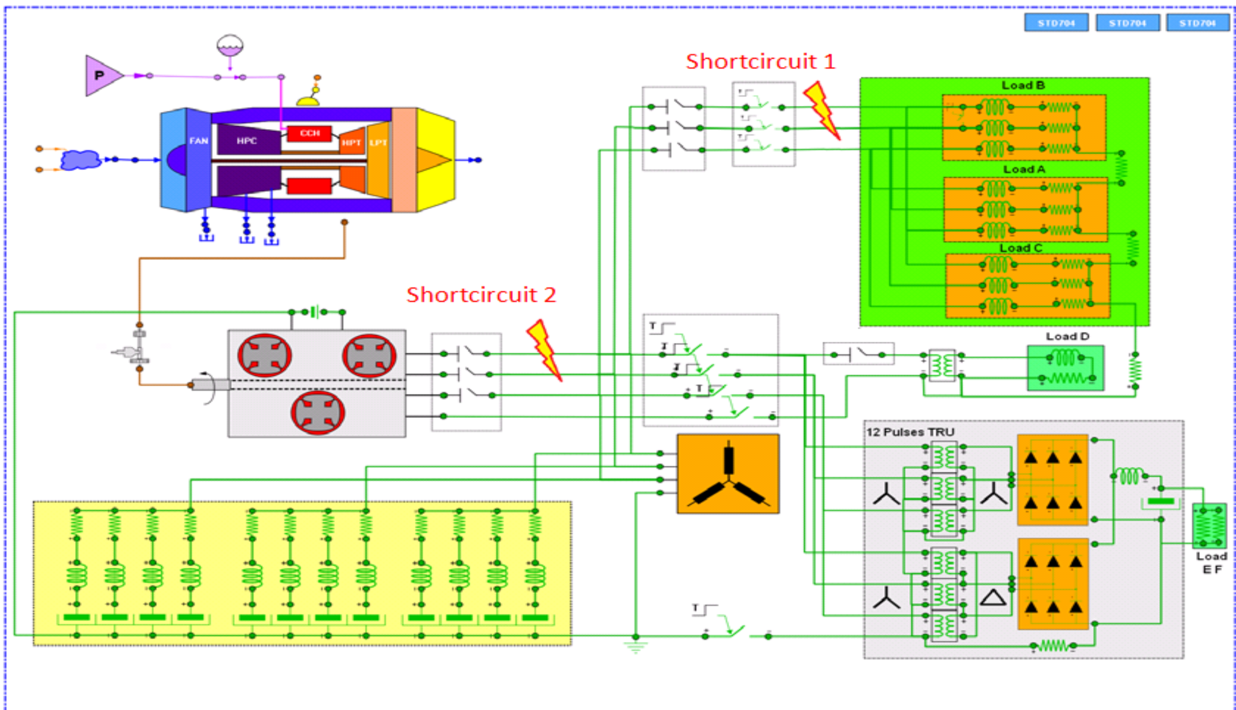

Fig 4.1: Advanced Aircraft Electrical Systems

CONCLUSION AND FUTURE SCOPE

Aeroplane is among the greatest inventions of the human mind and is indeed a complex engineering marvel. Its sheer enormity and the diversity of its components make the study of an aircraft and its operating principles a long and difficult journey. The time constraints under which this industrial training was pursued made it even more challenging. Nonetheless, I hope that however much could have been studied in this time; I have been successful in doing so. Being able to get a basic knowledge of the electrical, electronics, radio, communication, mechanical, instrumental aspects of an aircraft has been highly satisfying and enriching for me.

As an Electronics and Communication Engineer, the Radio Overhaul Shop provided me with the most valuable insights into an aircraft`s inner operations and I can say without any doubt in my mind that the scope which this single aspect of an aeroplane offers in terms of studying and understanding, is huge. I hope that I get opportunities in future to further my knowledge in this fascinating stream of physics.

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